

THANH LONG LY

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🌐 Thanh Long Ly

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Professional Experience

Jun. 2025 – Present

📌 **AI Engineer Intern**, Alpha Data Academy, Hanoi, Vietnam.

- Developed an **Agentic RAG System** for a leading manufacturer, architecting a **hybrid retrieval engine** (dense + sparse + RRF reranking) on **Qdrant** to improve search precision.
- Built scalable data ingestion pipelines and optimized LLM response alignment with business intents; deployed on **Render** with **Opik** monitoring and **MongoDB** logging.

Jan. 2025 – Present

📌 **AI Research Assistant**, Predictive Models and ML Research Lab, NEU.

- Specialized in fine-tuning transformer architectures (**BERT/RoBERTa**) for domain-specific NLP tasks, bridging the gap between academic research and industry applications.
- Delivered technical training sessions on machine learning workflows, fostering a collaborative engineering environment.

Sep. 2024 – Mar. 2025

📌 **Data Researcher**, Scientific Research Competition, NEU.

- Implemented **spatial econometric models with panel data** to quantify policy impacts, utilizing rigorous statistical validation for high-precision insights.
- Awarded the “**Student Research Award 2025**” for excellence in data-driven methodology and analysis.

Featured Projects

Oct. 2025 – Dec. 2025

📌 **Domain-Specific Conversational AI** (*Champion, Master AI Hackathon 2025*).

- Architected a **hybrid agentic system** (**LangGraph** + Retion.ai) with custom REST tools for ROI calculation, solving terminology ambiguity in B2B/B2C sectors.
- Won **1st Place** out of 20+ teams for delivering a high-precision QA solution with scalable infrastructure.

Jan. 2026 – Present

📌 **Real-Time Multimodal AI for IELTS Assessment**.

- Developed a low-latency (<500ms) voice agent using **Gemini 2.5 Flash Native** (WebSockets) for natural human-AI interaction.
- Built an automated scoring engine based on **IELTS band descriptors**, integrating real-time PCM audio streaming via a **React/Node.js** architecture.

Featured Projects (continued)

Oct. 2025 – Nov. 2025

■ Meteorological Time-Series Forecasting.

- Engineered a robust **Sklearn pipeline** and optimized 29 models (CatBoost, LightGBM, XGBoost) using **Optuna**, achieving **R^2 0.907**.
- Automated model tracking and deployment via **ClearML** and **FastAPI**, ensuring reproducible and scalable ML workflows.

Technical Skills

- **Programming:** Python (Expert), C++, SQL, JavaScript (React/Node.js).
- **AI/ML Stack:** PyTorch, TensorFlow, HuggingFace, LangChain, LangGraph, Scikit-learn, Optuna.
- **Data & Infra:** Qdrant (Vector DB), PostgreSQL, MongoDB, Supabase, Redis.
- **Tools & DevOps:** Docker, Kubernetes, AWS, Git, MLFlow, ClearML, FastAPI, Render.
- **Languages:** English (Professional), Vietnamese (Native), Chinese (Elementary).

Education

Aug 2023 – Aug 2027

■ National Economics University (NEU), Hanoi, Vietnam.

Major: **Data Science** in Economics and Business.

CGPA: 3.55/4.0

Core Coursework: Machine Learning, Data Structures & Algorithms, Econometrics, Discrete Math.